

KUSHAL KUMAR PELLURU

Software Engineer · Full Stack · Data Engineering · AI/ML

Bengaluru, India • pellurukushalkumarreddy@gmail.com • +91 9110363711 • linkedin.com/in/kushal-reddy-826a01255 • github.com/pellurukushalkumarreddy-png

PROFESSIONAL SUMMARY

Final-year Computer Science (AI) student at MAHE with 3 internships across enterprise AI, pharma automation, and machine learning. Built production-grade systems including an Agentic AI evaluation pipeline at Eli Lilly, a compliance automation tool at Aurobindo Pharma, and PharmaVerse — a full-stack pharma intelligence platform with 4 integrated tools powered by real APIs and Claude AI. Comfortable across the full stack (Python, React, Flask, AWS) and driven by shipping things that actually work.

TECHNICAL SKILLS

Languages	Python, JavaScript (ES6+), TypeScript, SQL, HTML/CSS, C, Embedded C
Frontend	React.js, Vite, Node.js, Express.js, Flask, REST APIs, D3.js, Recharts
AI / ML	TensorFlow, Keras, Scikit-learn, XGBoost, LangGraph, Agentic AI, Claude AI (Anthropic API), CNNs, DNN
Data & DB	Pandas, NumPy, PostgreSQL, MySQL, SQLAlchemy, Power BI, python-docx
Cloud & DevOps	AWS (Lambda, S3, RDS, ECS, IAM, VPC, CloudWatch), Vercel, Cloudflare Workers, GitHub Actions, Docker
APIs & Tools	Git, Jupyter, BeautifulSoup, Trafilatura, Jira API, Smartsheet API, ClinicalTrials.gov API, ChEMBL API, openFDA API, PubMed MCP

EXPERIENCE

Tech Intern — LRL (Agentic AI & Data Engineering) | Eli Lilly and Co. | Bengaluru, IndiaJan 2026 – Present

- Wrote Python-based evaluation pipelines for Taskly's ARIA use case using LangGraph — parsed AI-generated workflow outputs, applied custom scoring logic, and pushed structured results to AWS CloudWatch for real-time monitoring
- Built a Python web scraper (requests, subprocess, curl + Trafilatura) extracting regulatory documents from 6 international health authority sites, bypassing corporate SSL interception and HTTP/2 failures that broke standard libraries
- Wrote Python scripts using the Jira REST API and Smartsheet SDK to pull issue data across 3 projects, compute AT-RISK scores from missing due dates and backlog age, and auto-update Smartsheet rows — flagging 7 at-risk issues with zero manual effort
- Tools: Python, LangGraph, AWS (CloudWatch, S3, ECS, IAM), Jira REST API, Smartsheet SDK, Trafilatura, curl, subprocess, JSON

Software Validation Automation Intern | Aurobindo Pharma | Hyderabad, IndiaJun 2025 – Aug 2025

- Used Python (Flask) to build a web server that dynamically loaded and executed test scripts on demand — capturing stdout, screenshots, and pass/fail results through a browser UI built with HTML/CSS/JS
- Wrote Python scripts using python-docx to auto-generate audit-ready Word reports from test results, and used Pandas to aggregate pass rates across runs and surface failures for compliance review
- Tools: Python, Flask, Pandas, python-docx, HTML/CSS/JS, clipboard monitoring tools

Machine Learning Intern | Feynn Labs | RemoteFeb 2025 – Apr 2025

- Used Python (Pandas, NumPy) to clean and engineer features from raw EV market datasets; trained Scikit-learn and XGBoost models for demand segmentation, evaluated with Matplotlib visualizations, and deployed via Flask on AWS EC2
- Tools: Python, Pandas, NumPy, Scikit-learn, XGBoost, TensorFlow, Matplotlib, Flask, AWS EC2

PROJECTS

PharmaVerse — Pharma Data Intelligence Platform · React, TypeScript, Claude AI, D3.js, Recharts, ClinicalTrials.gov API, ChEMBL API, openFDA API, PubMed MCP, Vercel

- Built a 4-tool pharma intelligence platform in React + TypeScript, modeled after Lilly's Taskly — each tool independently consumes a real-world API and renders data-driven UI, deployed as a single product on Vercel

- TrialLens: async fetch calls to ClinicalTrials.gov API (400K+ trials) → Recharts dashboards for status, phase, sponsor, and enrollment trends | DrugNexus: D3.js force-directed graph mapping 20+ drugs × 22 targets × 173 diseases from ChEMBL API
- BioInsight: PubMed MCP queries 36M+ articles live; abstracts piped into Claude AI API (Anthropic Messages API) for plain-language summarization and research Q&A | FDAForge: TypeScript ETL pipeline on openFDA API normalizing nested JSON across 29K+ drugs into Recharts analytics

Atlas — Personal Career & Learning Platform · *React, Claude AI (Anthropic API), Node.js, Cloudflare Workers, Adzuna API, Remote API*

- Built a React + Node.js app with 5 modules (learning tracker, job tracker, resource library, study plans, AI mentor 'Sage'); integrated the Anthropic Messages API to deliver personalized career guidance using user context as a system prompt
- Wrote a Cloudflare Worker (JavaScript) on a cron schedule — daily fetches role-matched jobs from Adzuna and Remote APIs, formats an HTML email digest, and delivers it at 7 AM IST with one-click 'Add to Atlas' tracking links

CKD Early Detection — ML Research (Journal Paper in Progress) · *Python, XGBoost, DNN, NSGA-II, Scikit-learn, TensorFlow, Pandas*

- Used Python (Pandas, Scikit-learn) to preprocess clinical CKD datasets and trained four models — XGBoost, SGD, TensorFlow DNN, and NSGA-II (multi-objective genetic algorithm optimizing accuracy vs. false-negative rate); evaluated with ROC-AUC, F1, and confusion matrix — targeting peer-reviewed publication

EDUCATION

B.Tech — Computer Science & Engineering (AI) | **Manipal Academy of Higher Education (MAHE)** | Manipal, India
Sep 2022 – May 2026

- GPA: 7.88 / 10.0 | Specialization: Artificial Intelligence & Machine Learning
- Relevant coursework: DSA, Machine Learning, Database Systems, Cloud Computing, Computer Vision, NLP, Software Engineering, UI/UX Design

CERTIFICATIONS & ACHIEVEMENTS

-
- Machine Learning for Computer Vision — MathWorks | Object Tracking & Motion Detection — MathWorks
 - Oracle SQL Databases — LearnQuest | Embedded Software & Hardware Architecture — Univ. of Colorado Boulder
 - Principles of UX/UI Design — Coursera | Cisco Forecast League — Achieved 90% accuracy in sales forecasting using ML
 - Road Safety Hackathon — Proposed AI-driven solution for real-time traffic violation detection